Flow Laws of Polyphase Aggregates from End-Member Flow Laws

TERRY E. TULLIS, FRANKLIN G. HOROWITZ[1], AND JAN TULLIS

Department of Geological Sciences, Brown University, Providence, Rhode Island

An incompressible finite element model has been used to study the plane strain deformation of two-phase aggregates deformed by dislocation creep. Input for the model includes the power law flow laws of the two end-member phases and their volume fractions and configuration. The model calculates the overall flow law of the aggregate as well as the stress and strain rate variations within it. The input flow laws were experimentally determined for monomineralic aggregates of clinopyroxene and plagioclase. Results were calculated for a temperature of 1000°C, strain rates from 10^4 to 10^{-12} s^{-1}, and stresses of 1-1000 MPa. For these conditions, the end-member flow laws intersect on a log stress versus log strain rate plot at 10^8 s^{-1}. Some runs were made on finite element grids fit to an actual diabase texture (~64% pyroxene, ~36% plagioclase.) Other runs were made on idealized geometries to test the effects of varying the volume fraction of two phases, shape of inclusions, and relative strengths of inclusion and matrix. Important results include the following: (1) The model results satisfy the requirement that the aggregate strength must lie between the bounds set by the end-member flow laws and those set by assumptions of uniform stress and uniform strain rate. (2) The calculated diabase flow law matches well with that experimentally determined. (3) The aggregate strength within the uniform stress and uniform strain rate bounds is primarily affected by volume fraction, although certain phase geometries can also affect the strength. (4) Although the flow law for an aggregate of power law phases need not be a simple power law, we find it to be a good approximation. We have developed two simple methods of estimating the strength of an aggregate, given the end-member flow law parameters and volume fractions; both give results that agree with the finite element model calculations. (1) One method takes into account the phase geometry and gives a strength for the aggregate at any strain rate. (2) The other method can be used even if the phase geometry is unknown and gives expressions for the aggregate flow law parameters.

INTRODUCTION

Knowledge of the ductile flow properties of both the crust and the mantle is important for understanding a wide range of geologic processes. Models of earthquake distributions [e.g., *Sibson*, 1982], extensional tectonics [e.g., *Zuber et al.*, 1986], and convection [e.g., *Christensen*, 1984] all require appropriate flow laws for the rocks in the regions of interest. At present, it is common practice to model crustal deformation with a "wet quartz" flow law, and upper mantle deformation using an olivine flow law [e.g., *Chen and Molnar*, 1983]. Although most crustal basement rocks are polyphase aggregates, there has been no way to evaluate the errors involved in approximating them as single-phase aggregates.

Flow laws must be determined in the laboratory and then extrapolated to natural conditions. Monomineralic aggregates experimentally deformed with axial symmetry in steady state dislocation creep exhibit a power law form of flow law [e.g., *Nicolas and Poirier*, 1976]:

$$\dot{\varepsilon} = A\,\sigma^{n}\,e^{-Q/RT} \qquad (1)$$

where $\dot{\varepsilon}$ is the axial strain rate, A is the pre-exponential factor, σ is the differential flow stress, n is the stress exponent (often close to 3), Q is the activation enthalpy, R is the gas constant, and T is the absolute temperature. The experimentally determined power law flow laws are usually extrapolated linearly (with a slope equal to n and an intercept determined by Q and A) on a log stress versus log strain rate plot [e.g., *AveLallemant and Carter*, 1970; *Heard and Raleigh*, 1972].

[1]Now at Department of Civil Engineering, Northwestern University, Evanston, Illinois.

Paper number 90JB02491.
0148-0227/91/90JB-02491$05.00

Experiments are insufficient to determine the rheology of polyphase aggregates at natural conditions if the flow laws for their constituent phases are different. First, the flow law for such an aggregate need not be a simple power law and thus may not be a straight line on a log stress versus log strain rate plot. Second, flow laws for the constituent phases will have different slopes on a log stress versus log strain rate plot and thus will intersect. For a two-phase aggregate the single intersection point (at a given temperature of deformation) may be called the crossover or equiviscous point (see example on Figure 3). Depending on the conditions of the equiviscous point, it is quite possible that the phase which is stronger at experimental conditions is weaker at natural conditions.

Recently, there have been a number of attempts to estimate the strengths of two-phase aggregates in terms of the strengths, proportions, and configuration of their constituents, with particular attention to how the configuration and thus the aggregate strength might vary with progressive strain. *Tharp* [1984] suggested that some aggregates with a strong phase forming an interconnected framework and a much weaker dispersed phase could be modelled as porous powder metals. Several experimental studies of natural and synthetic two-phase aggregates have determined how strength varied as functions of volume proportion of the weak phase and of strain [*LeHazif*, 1978; *Dell'Angelo and Tullis*, 1982; *Price*, 1982; *Tullis and Mardon*, 1984; *Jordan*, 1987; *Ross et al.*, 1987]. Recently, several theoretical studies have addressed these same problems [*Jordan* 1988; *Handy*, 1990; *Gilotti*, 1990]. These studies have emphasized the importance of the weak phase (or matrix) in controlling the aggregate strength and have discussed how its amount or distribution may change with increasing strain. In particular, *Handy* [1990] has suggested that polyphase aggregates can be approximated as belonging to one of three basic classes: (1) a load-bearing framework of a strong phase with a relatively small volume proportion of weak phase (<~25%), in which case the strength of the aggregate approaches that of the strong phase; (2) isolated clasts of a strong phase dispersed in a much weaker matrix (strength contrast >10/1), in which case the strength of the aggregate approaches that of the weak phase; and (3) two phases which have a relatively low strength contrast (<~10/1), in which case both will undergo extensive flow and the strength of the aggregate will lie between those of the end-members.

None of the above studies have attempted an exact analytical solution to the deformation of polyphase aggregates by dislocation creep. This is a difficult problem because of the nonlinear rheology of the individual phases as well as the complex and variable phase configurations shown by real rocks. The only analytic technique for this type of nonlinear problem found in the literature [*Chen and Argon*, 1979] is limited to deformation of spherical inclusions (or circular cylindrical inclusions in plane strain) in an infinite matrix and makes an untested assumption of strain homogeneity in the inclusions. We have chosen an incompressible two-dimensional finite element model [*Horowitz et al.*, 1981] because it allows a numerical solution of this mixed nonlinear rheology deformation problem, using bodies ("grains") of variable rheology and configuration

In this paper we report the results of a finite element modeling study of two-phase aggregates, using as input the experimentally determined flow laws for plagioclase and clinopyroxene [*Shelton*, 1981; *Shelton and Tullis*, 1981]. We have modelled the deformations for samples with a diabase texture and have compared the results with an experimental flow law for diabase [*Shelton*, 1981]. We have also investigated the effects of systematic changes in the volume proportions and configurations of the phases for various idealized textures. The model calculations have allowed us to develop and test a simple method for calculation of the flow law for a polyphase aggregate, based on the flow laws and volume fractions of its constituents, which appears to provide quite accurate approximations. This allows one to predict the strengths of a large number of rocks at geologic conditions based on the experimentally determined flow laws of a relatively small number of monomineralic aggregates.

THEORETICAL BOUNDS ON AGGREGATE STRENGTH

What constraints can be put on flow laws appropriate to the crust and mantle? On a log stress versus log strain rate plot the flow law for a polyphase aggregate is constrained to lie somewhere between the limits set by the flow laws of the constituent phases and to pass through the equiviscous point. This places some constraints on the strength of the aggregate at natural conditions: close constraints if the constituents have nearly the same flow law and poorer constraints if they do not. In fact, closer bounds on the aggregate strength can be placed by taking the volume fractions of the phases into account.

The behavior of power law aggregates is analogous to the behavior of linear elastic aggregates [e.g., *Hashin*, 1964]: at a given strain rate an upper stress bound is provided by the assumption of uniform strain rate within the aggregate, and a lower stress bound is provided by the assumption of uniform stress. These bounds are functions of the volume fractions of the phases. In the case of elasticity of single-phase polycrystalline aggregates the uniform strain and uniform stress bounds are called the Voight and Reuss bounds, respectively, and they are determined from the single-crystal elastic constants. Neither the uniform strain rate (or strain) nor the uniform stress bound is physically possible, because in the former case the stresses across phase boundaries would be discontinuous, whereas in the latter case strain rate (or strain) compatibility would not be satisfied. Nevertheless, *Hill* [1965] has shown that these two unrealistic situations represent rigorous bounds to the actual stress supported by the aggregate for the linear elastic situation.

The uniform stress and strain rate bounds also represent rigorous bounds in our case of the non-linear flow of a mixture, as can be seen by considering the following argument. If the stress for the aggregate flow law were outside the uniform strain rate bound, the stronger phase would on average exhibit a larger strain rate than the weaker phase, whereas if the stress for the aggregate flow law were outside the uniform stress bound, then the weaker phase would on average support a higher stress than the stronger phase. Both of these cases are physically impossible; they would result in a higher energy dissipation rate than if the aggregate were deforming with a uniform strain rate or a uniform stress, and thus the latter cases represent rigorous bounds. Even narrower rigorous bounds than these may exist, analogous to the Hashin-Shtrikman bounds for polyphase mixtures of elastic materials [*Hashin and Shtrikman*, 1963], but we have not discovered them.

To find the uniform strain rate and uniform stress bounds, begin with two flow laws of the form of equation (1) in which the pre-exponential factor A and the temperature dependence $e^{-Q/RT}$ have been combined into a quantity B:

$$\dot{\varepsilon}_1 = B_1 \sigma_1^{n_1} \tag{2}$$

$$\dot{\varepsilon}_2 = B_2 \sigma_2^{n_2} \tag{3}$$

where the subscripts denote quantities for phases 1 and 2. The uniform strain rate solution is found by equating both $B_1 \sigma_1^{n_1}$ and $B_2 \sigma_2^{n_2}$ to the desired uniform strain rate $\dot{\varepsilon}$ and solving these two equations for σ_1 and σ_2 in terms of $\dot{\varepsilon}$. The aggregate stress is then found, using the principle of virtual work (equation (7)), by taking the volume average of these stresses:

$$\sigma = f_1 \sigma_1 + f_2 \sigma_2 \tag{4}$$

$$\sigma = f_1 (\dot{\varepsilon}/B_1)^{\frac{1}{n_1}} + f_2 (\dot{\varepsilon}/B_2)^{\frac{1}{n_2}} \tag{5}$$

where f_1 and f_2 are the volume fractions of the two phases. The uniform stress solution is found by substituting the unknown uniform stress σ for σ_1 and σ_2 in (2) and (3) and finding the average strain rate by taking the volume average of the strain rates from (2) and (3) in a manner similar to (4). Such an equation gives the strain rate in terms of the stress. In order to find the stress, given the strain rate, this equation can be rearranged to look like:

$$\sigma = [\dot{\varepsilon}/(f_1 B_1 + f_2 B_2 \sigma^{n_2 - n_1})]^{\frac{1}{n_1}} \tag{6}$$

This expression can be solved iteratively for σ and converges rapidly; the uniform strain rate stress from (5) can be used as a first estimate for the σ on the right side of (6).

In subsequent sections we will compare the results of our finite element calculations with these theoretical bounds. Relevant plots are presented in Figures 3, 6, and 9*a*. Figure 9*a* illustrates the dependence of the uniform stress and strain rate bounds on volume fraction; all these figures illustrate how these bounds place a more stringent constraint on the strength of the aggregate than do the end-member bounds.

APPROACH AND TECHNIQUES

Finite Element Model

The particular incompressible power law finite element formulation used in this study was adapted from one developed by *Needleman and Shih* [1978]. The program satisfies the incompressibility constraint by employing quadrilateral elements arranged in strips extending from one boundary of the body to the other. Each quadrilateral is subdivided by its diagonals into four constant strain triangles. The resulting configuration of nodes allows the determination of some nodal displacements from others using the incompressibility constraint. Arbitrary boundary conditions cannot be used at the ends of the strips, but most common boundary conditions are acceptable. The scheme results in a stiffness matrix with a sufficiently narrow bandwidth that it is computationally tractable. Elemental strain rates and stresses are derived from a knowledge of the nodal velocity field and the material rheologies. We have modified this program to allow for the deformation of multiple phases, each with a different set of material constants.

The key features of our numerical simulations are as follows. The array of quadrilateral elements is deformed in two-dimensional plane strain. "Free slip" boundary conditions on the top and bottom surfaces of the sample constrain them to remain planar and parallel, and the sides are specified to be stress-free. Each deformational increment is carried out by displacing the top boundary a fixed amount toward the bottom boundary. Thus the body deforms as if it were a column, infinite in the Y direction, composed of endlessly repeating mirror symmetric pairs of the sample geometry. The computations are carried out for a unit time increment (1 s), so the above conditions correspond to a constant strain rate shortening, the magnitude of which is given by the imposed total strain.

The strains achieved in this model deformation are extremely small (10^{-4} to 10^{-12}). We have not used the program to compute large strains and the resultant changes in phase configuration because this would be prohibitively expensive. However, we have used the phase geometry of a diabase sample experimentally deformed to a large strain as one of our input cases for the finite element modeling and we have compared the results to those for an undeformed diabase texture.

Experimental flow laws are determined in axisymmetric deformation whereas the model behavior is determined in plane strain deformation. Thus we have converted the experimental flow laws for the monomineralic aggregates into plane strain equivalents in order to serve as input for the model. We have also converted the experimental flow law for diabase into a plane strain equivalent in order to compare it with the model predictions. In order to make these conversions, the pre-exponential factor A of equation (1) must be multiplied by a scale factor equal to $1/2\ (3^{(n+1)/2})$. This scale factor arises from the constraints upon the strain rate tensor given by the two types of constant volume deformation, and assumes that the proper generalization of equation (1) is that which relates the second invariant of the strain rate tensor to the second invariant of the deviatoric stress tensor raised to the power n [*Nye*, 1953; *Stocker and Ashby*, 1973, Appendix 2]. The plane strain equation that results from use of this scale factor relates a given component of the strain rate tensor to the same component of the deviatoric stress tensor.

Because the system studied is mathematically nonlinear, the solution must be found by an iterative process. As a practical matter, finding and accepting solutions that converged was very time consuming. All of the results reported in this paper were obtained using a fairly involved convergence scheme. One of the original developers of the program, A. Needleman (personal communication, 1982), believes these results do indeed demonstrate convergence, although he cautions that multiple power law phases could conceivably produce "multiple" solutions. We believe that this has not happened because in several test cases the same solution was reached from two different convergence schemes.

Our model deformations were performed at a temperature of 1000°C, constant strain rates of 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}, and $10^{-12}s^{-1}$ and aggregate flow stresses of 1 to 1000 MPa. The temperature is somewhat high for applications to the crust, but it is within both the experimental and the natural range, and it is especially useful for our purposes because it results in an equiviscous point midway in our chosen strain rate range (Figure 3). The strain rates were chosen to span the range from experimental to natural. The stresses are admittedly high, as were those used in the experimental flow law determinations (100-600 MPa), but can be justified here because the main purpose of our study is not to gather detailed information about the natural flow of diabase but to formulate general principles about the flow behavior of two-phase aggregates.

A wealth of information results from each model deformation increment, because the program calculates the stress and strain rates for each element. We have used three principal types of output in this study. First, direct output of the stresses and strain rates is extremely useful for seeing the heterogeneities that can exist within and between phases of a polyphase aggregate. Second, we were primarily interested in the aggregate flow stress, and we have calculated this from the elemental stresses and strain rates by using the principle of virtual work:

$$\int_s \tau_i u_i \, dS = \int_v \sigma_{ij} \dot{\varepsilon}_{ij} \, dV \qquad (7)$$

In the present context, this principle relates the externally applied power (the inner product of traction τ_i and velocity u_i vectors over the bounding surfaces S) to the mechanically dissipated internal power (the inner product of the stress σ_{ij} and strain rate $\dot{\varepsilon}_{ij}$ tensors over the volume V of the body). From a knowledge of the stress and strain rate for each element, the volume (area times unit depth) of each element, and the velocity boundary conditions, an average surface traction can be calculated from equation (7). Given the velocity boundary conditions imposed in our "samples," these tractions reduce to a single vector-component normal to the top and bottom surfaces. This value is twice the desired average deviatoric flow stress for the aggregate (since the sides are stress-free).

A third useful output is the "stress working power" (i.e., "energy dissipation rate" or "power dissipation"), a scalar field derived by integrating the right-hand side of (7) over each element instead of the entire body. This scalar is a measure of each element's contribution to the overall energy dissipation rate (and, given our boundary conditions, to the aggregate stress) imposed on the deforming aggregate. Output of this quantity is useful for determining the heterogeneity of energy dissipation for different configurations (shapes and arrangements) of the phases. In our figures we have chosen to display the power dissipation per unit volume for each grain, so that results for grains with different sizes can be more easily compared.

Input Parameters

In all of our calculations we have used as input the power law flow laws experimentally determined in axial compression for plagioclase and clinopyroxene by *Shelton* [1981], and we have tested the model predictions against his experimentally determined flow law for diabase. These flow laws in axisymmetric form are

$$\text{Plagioclase} \quad \dot{\varepsilon} = 1.18 \times 10^{6}\ s^{-1}\ GPa^{-3.9}\ \sigma^{3.9}\ e^{-\frac{234\ kJ/mol}{RT}} \qquad (8a)$$

$$\text{Pyroxene} \quad \dot{\varepsilon} = 9.95 \times 10^{8}\ s^{-1}\ GPa^{-2.6}\ \sigma^{2.6}\ e^{-\frac{335\ kJ/mol}{RT}} \qquad (8b)$$

$$\text{Diabase} \quad \dot{\varepsilon} = 2.58 \times 10^{6}\ s^{-1}\ GPa^{-3.4}\ \sigma^{3.4}\ e^{-\frac{259\ kJ/mol}{RT}} \qquad (8c)$$

where differential stress is in units of GPa. These same flow laws, converted to plane strain form and evaluated at our run conditions of 1000°C, are

$$\text{Plagioclase} \quad \dot{\varepsilon} = 2.18 \times 10^{-3}\ s^{-1}\ GPa^{-3.9}\ \sigma^{3.9} \qquad (9a)$$

$$\text{Pyroxene} \quad \dot{\varepsilon} = 6.48 \times 10^{-5}\ s^{-1}\ GPa^{-2.6}\ \sigma^{2.6} \qquad (9b)$$

$$\text{Diabase} \quad \dot{\varepsilon} = 3.42 \times 10^{-4}\ s^{-1}\ GPa^{-3.4}\ \sigma^{3.4} \qquad (9c)$$

The second type of necessary input for the modeling concerns the volume proportions and the configuration of the two phases. We have taken two approaches here: some model deformations use the phase boundaries traced from actual diabase thin sections, whereas others use various idealized geometries with a range of different volume proportions. The diabase texture represents the case where both phases are almost equally interconnected. In order to reproduce the texture of the Maryland diabase a photograph was taken of a thin section, and a grid of elements was hand-fit to a representative area (approximately 50 grains) of the photo. The volume proportions of the grids agreed to within ±2% with that of the actual diabase. This procedure was followed for two separate thin sections of undeformed Maryland diabase and for one section of a sample experimentally shortened 45% at 1500 MPa confining pressure, 1000°C, and $10^{-4}s^{-1}$. Examples of the grids for both the undeformed and the deformed samples are shown in Figure 1. The grid size was chosen so as to have at least one quadrilateral element per mineral grain, while remaining computationally affordable.

We ran a test case for one of the diabase textures in which each normal size element was subdivided into four smaller ones (Figures 4*c* and 4*d*). This fine grid was then deformed at a strain rate of $10^{-4}s^{-1}$ to compare with the results for a normal grid at the same conditions. The aggregate flow stresses for the normal and fine grids match to within 4%, and the variations of stress and strain rate in the two sizes of grids are similar (Table 1 and Figure 4). The flow stress for this fine grid has been added to the plot in Figure 6c to facilitate comparison. Inspection of Figure 4 shows that the stress and strain rate fields for the fine grid compare favorably with those for the normal grid. With these data, we feel confident that our normal density grids provide information which is sufficiently accurate for the purpose of this study.

A number of idealized textures were also investigated in this study. We wished to determine general principles concerning the dependence of the strength of a polyphase aggregate upon its texture and volume proportions, including textures consisting of one continuous phase and one dispersed phase, and it is not easy to find natural rocks actually showing these systematic variations in order to digitize them directly. Thus we have used idealized textures consisting of one continuous phase and one dispersed phase for which we can vary the volume proportions while leaving the configuration substantially unchanged and, conversely, for which we can vary the configuration while leaving the volume proportions unchanged. In order to test the effect of angularity we have done calculations for both circular and square inclusions of one phase set in a matrix of the other phase. With the boundary conditions we used, the geometry is equivalent to that in Figure 2. An additional benefit of testing such geometries is that one of them somewhat resembles the geometry of a circular inclusion in an infinite matrix for which *Chen and Argon* [1979] obtained an analytical solution. For each of the simple geometry models, only one quarter of each unit cell shown in Figure 2 was used in the computation due to the aforementioned symmetry. The density of the finite element grid within that quarter unit cell is shown in Figures 7 and 8.

Assumptions and Limitations

There are several assumptions involved in our choice of input flow laws for the two-phases:

1. We assume that each material follows the same flow law over the whole range of conditions chosen for the modeling. This assumption is not inherent in the finite element method, but it does simplify the modeling.

2. We assume that there are no interactions between the two phases in the polyphase aggregate fundamentally different from those in monomineralic aggregates of the end-members.

3. The model is assumed to represent steady state dislocation creep, just as the experimentally determined input flow laws were assumed to represent steady state. It does not explicitly model changes in mechanical behavior that may result from textural evolution with increasing strain, including changes in phase size, shape, or configuration; reactions to form new phases; a switch in dominant deformation mechanism (as may accompany grain size reduction by syntectonic recrystallization); and development of a crystallographic preferred orientation. Thus in order to determine how the aggregate flow law changes with strain, one must determine experimentally or from observations of naturally deformed rocks what phase shapes and configurations are present and what deformation mechanisms are dominant at low, intermediate, and high strain at the conditions of interest and then compute model predictions for these different situations.

4. We assume that the two specific monomineralic aggregates experimentally studied are appropriate "end-members" for the diabase experimentally studied and against which the model predictions have been tested as well as for naturally deformed plagioclase-pyroxene aggregates. The diabase experimentally studied (Maryland diabase) is composed of 50-70% clinopyroxene ($En_{55}Wo_{12}Fs_{33}$) and 30-50% plagioclase ($Ab_{27}An_{71}Or_2$) with <2% accessory opaques. The Sleaford Bay clinopyroxenite is composed entirely of salite ($En_{40}Wo_{50}Fs_{10}$). *Shelton* [1981] determined the flow law for an anorthosite of composition $Ab_{25}An_{75}$, very close to the

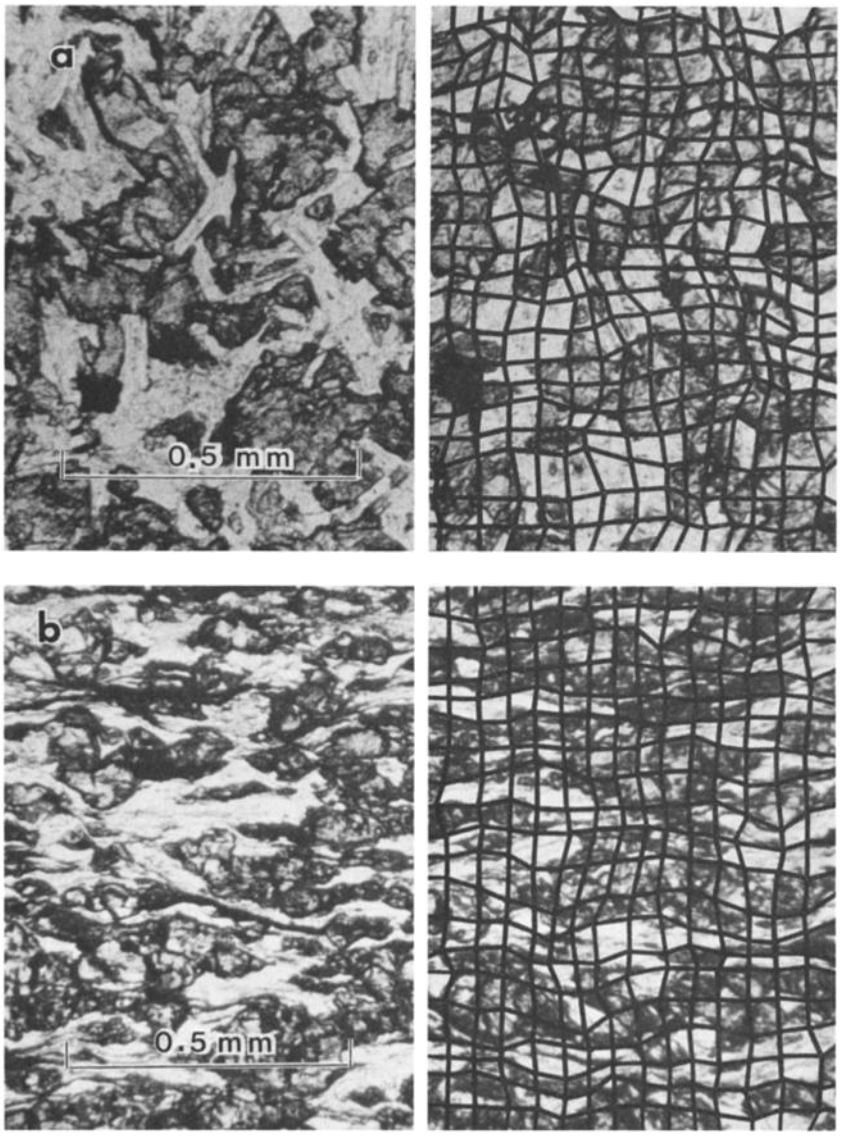

Fig. 1. Photomicrographs of the textures of (*a*) undeformed and (*b*) experimentally deformed Maryland diabase showing the superimposed finite element grids used in modeling (grids 2 and 3, respectively). The light areas are plagioclase (35%), and the dark areas are clinopyroxene (65%).

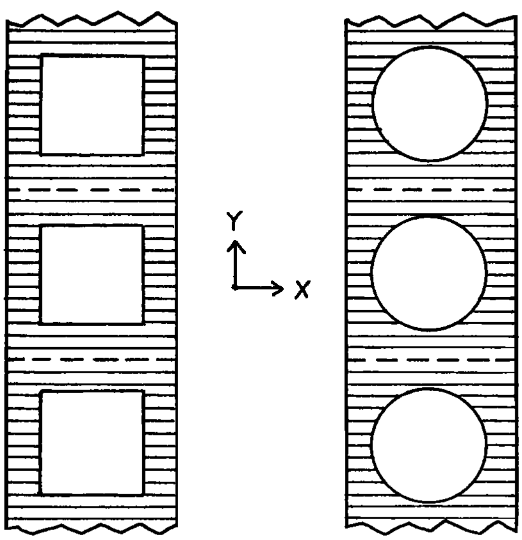

Fig. 2. Illustration of two of the four idealized geometries tested. The unshaded material represents plagioclase and the shaded material represents clinopyroxene, as in all other figures. The other two idealized geometries tested have clinopyroxene as the inclusion. This figure shows several unit cells to illustrate the boundary conditions: sides (parallel to Y) which are stress-free and top and bottom boundaries (parallel to X) on which the normal displacement is uniform and the shear stresses are zero.

composition of the plagioclase of the diabase. However, because he did far more extensive work on a polycrystalline albite rock ($Ab_{96}An_3Or_1$) and found no apparent difference in the flow laws for these two compositions, we have used his flow law for albite. Optical and transmission electron microscopy indicate that the microstructures are almost identical for the two monomineralic aggregates and the two phases in the diabase deformed at the same conditions. This indicates that the same deformation mechanisms and flow laws were involved [*Kronenberg and Shelton*, 1980].

There are several limitations to this model:

1. The model neglects crystal anisotropy; it assumes that each grain deforms with the average properties of an aggregate of that phase, although few silicates have five independent slip systems. Crystal anisotropy will increase the spatial variability of local stresses and strain rates. However, our input flow laws are for monomineralic aggregates and thus already embody the effects of crystal anisotropy. If the stress and strain rate inhomogeneity in a polyphase aggregate does not significantly differ from that in the end-member monomineralic aggregates, then the neglect of crystal anisotropy will not introduce much error. However, if one phase has few independent slip systems, then its deformation may be easier in a polyphase aggregate where inhomogeneous deformation may be allowed by the other phase(s).

2. This continuum model does not allow grain boundary sliding; all element boundaries are "welded" to their nearest neighbor by sharing of a common node.

3. The model results in this paper were calculated for pure shear, but they suggest certain generalizations about differences that would be observed for simple shear. The configuration of the phases is the most important factor for causing differences in flow strengths between pure and simple shear deformation of polyphase aggregates. If there is any tendency for a layering or foliation, then pure shear normal to the foliation will result in greater strengths than simple shear parallel to it [e.g., *Jordan*, 1987]. In simple shear it is more likely that deformation will be localized in the weaker matrix and the aggregate will deform at a lower stress, close to the uniform stress bounds. This would be the case for the geometry of Figure 2, for example, if the imposed boundary conditions consisted of a shear displacement parallel to the x axis rather than the compression parallel to the y axis that we have studied. An aggregate strength close to the uniform stress bound would similarly be expected for pure shear at 45° to a foliation, although with progressive strain and rotation of the foliation the strength should increase and move away from this bound.

4. The polyphase aggregate flow laws predicted by the model depend on the quality of the input flow laws. At present there are unknown and possibly large errors in the extrapolations of dislocation creep flow laws determined in solid media apparatus [e.g., *Paterson*, 1987; *Tullis and Yund*, 1989], due to uncertainties in the flow stresses, use of high flow stresses, inclusion of more than one deformation mechanism, questionable achievement of steady state, and the unknown pressure dependence of hydrolytic weakening. However, the recent development of the molten salt assembly by *Green and Borch* [1989] should enable flow laws to be more accurately determined at high pressure in the near future.

RESULTS

In this section we present our results for the strengths and flow laws of the actual and idealized polyphase aggregates and the variations in stress, strain rate, and power dissipation within the aggregates. Most of the discussion of these results, including generalizations concerning the flow laws of polyphase aggregates and the effects of volume fractions and phase configuration, is presented in the Discussion section below.

Diabase Textures

For the three diabase grids, model calculations of the aggregate flow stress made at 1000°C at five different constant strain rates from 10^{-4} to $10^{-12} s^{-1}$ are shown in Figure 3 on a plot of log stress versus log strain rate, together with *Shelton's* [1981] experimentally determined flow law for Maryland diabase (converted to plane strain form). For the model

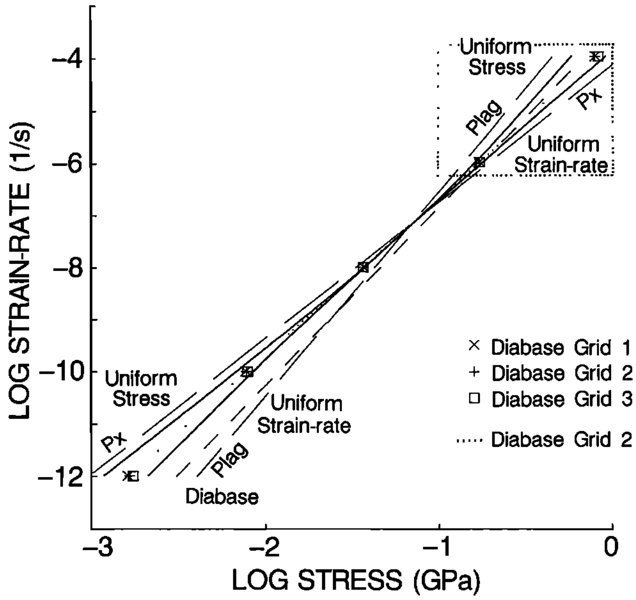

Fig. 3. Plot showing computed and experimental plane strain flow laws at a temperature of 1000°C, including (1) the experimentally determined flow laws for the plagioclase and pyroxene single phase aggregates (long-dashed lines) and for diabase (short-dashed lines), (2) the computed flow laws for the three diabase grids (symbols and dotted line), and (3) the uniform stress and uniform strain rate bounds for plagioclase-pyroxene aggregates (upper and lower solid lines, respectively). The dotted rectangle shows the conditions over which experimental data were collected. Note that pyroxene is stronger at experimental strain rates but at about $10^{-7} s^{-1}$ and 1000°C there is an equiviscous point and at slower strain rates the phase strengths reverse due to their differing stress exponents. An expanded view of the data points is shown in Figures 6*b* and 6*c*, together with data for the idealized geometries.

calculations the strain rates plotted are the average maximum shortening strain rates for the aggregate, and the stresses are the maximum deviatoric compressive stresses for the aggregate (half the average traction applied to the top or bottom surface of the model sample shown in Figure 4).

Examples of the output stresses and strain rates for the individual elements for one of the undeformed diabase grids are shown in Figures 4*a* and 4*b*, respectively. Figure 4*a* shows the magnitudes and orientations of the maximum principal deviatoric compressive stress; they are higher in the pyroxene grains, the stronger phase at this strain rate ($10^{-4} s^{-1}$). The local

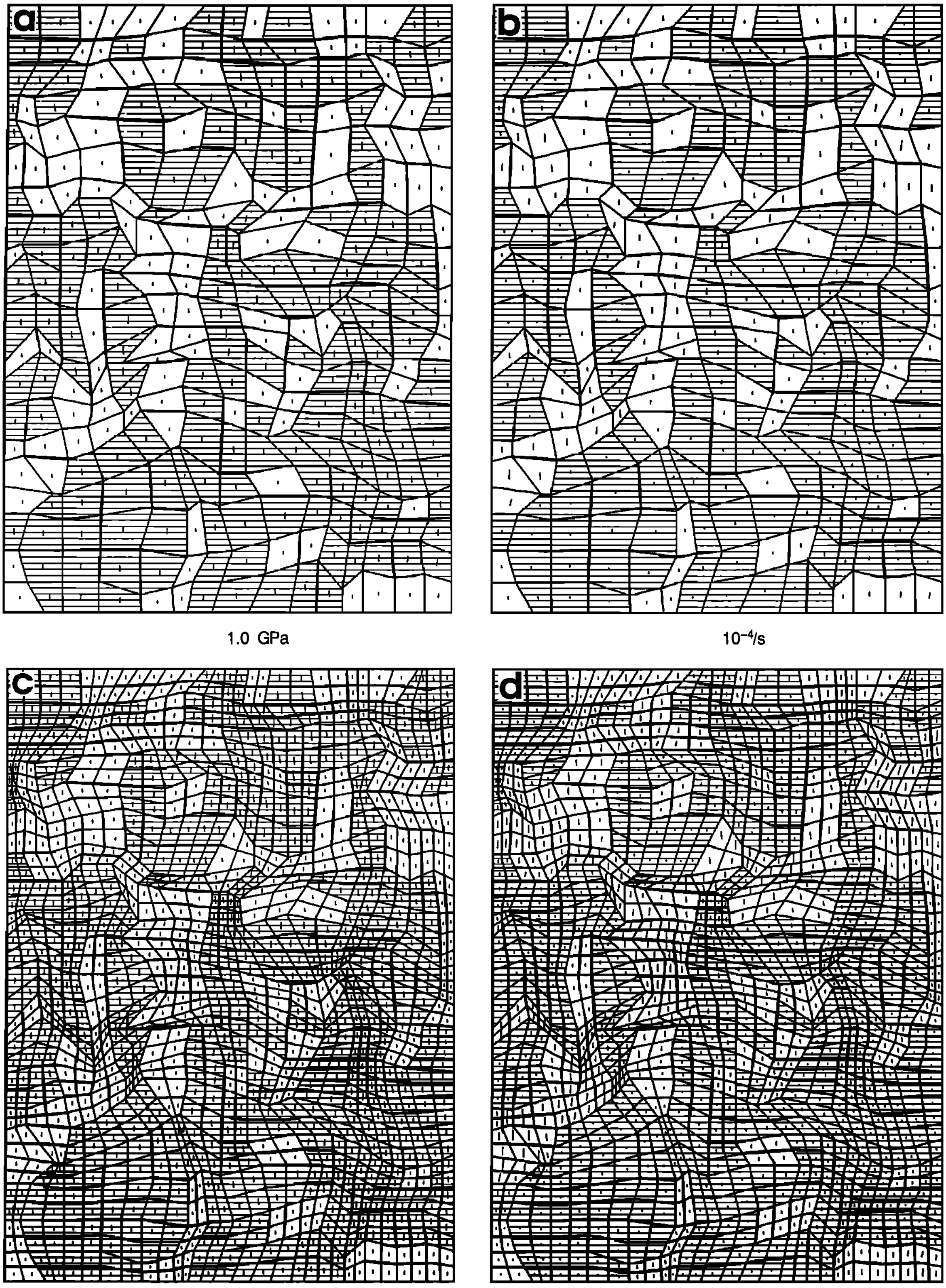

Fig. 4. Stress and strain rate distributions for grid 1, of standard size (Figures 4*a* and 4*b*) and fine size (Figures 4*c* and 4*d*), at $10^{-4}s^{-1}$ and 1000°C where pyroxene (shaded) is the stronger phase. Figures 4*a* and 4*c* show the orientations and magnitudes of the maximum principal compressive stress; Figures 4*b* and 4*d* show the orientations and magnitudes of the maximum principal deviatoric compressive strain rate. Scale bars for these are indicated; the same scales apply to both grids.

deviatoric stresses range from a maximum of 1.83 times (and an average of 1.27 times) the aggregate deviatoric stress in the pyroxene grains, to a minimum of 0.46 times (and an average of 0.61 times) the aggregate stress in the plagioclase grains (Table 1). Figure 4b shows the corresponding magnitudes and orientations ot the maximum principal shortening strain rates; they are higher in the plagioclase grains, the weaker phase at this strain rate. The strain rates range from a maximum of $3.14x10^{-4}s^{-1}$ in the plagioclase to a minimum of $0.25x10^{-4}s^{-1}$ in the pyroxene (Table 1). Generally speaking, the variation

TABLE 1. Stress and Strain Rate Averages and Ranges for Both Phases in Normal and Fine Versions of Grid 1 at $10^{-4}s^{-1}$

Phase	Ratio of Phase to Aggregate Deviatoric Stress*		Strain Rate, $10^{-4}s^{-1}$	
	Normal Grid	Fine Grid	Normal Grid	Fine Grid
Pyroxene				
Average ± s.d.	1.27 ± 0.19	1.27± 0.22	0.78 ± 0.31	0.70 ± 0.33
Range of Values	0.82 - 1.83	0.76 - 1.92	0.25 - 1.91	0.17 - 2.00
Plagioclase				
Average ± s.d.	0.61 ± 0.06	0.64 ± 0.08	1.49 ± 0.57	1.66 ± 0.73
Range of Values	0.46 - 0.76	0.41 - 0.82	0.47 - 3.14	0.27 - 4.15

*Aggregate deviatoric stress for normal grid = 804 MPa. Aggregate deviatoric stress for fine grid = 774 MPa.

between the two phases is not large; the average values of the deviatoric stresses and the strain rates in the two phases are within a factor of 2 of each other. Because of the nonlinearity of the flow law, the overlap between the pyroxene and the plagioclase strain rates is greater than that between their stresses (Table 1).

The power dissipation density gives an indication of the overall heterogeneity of behavior in the aggregate. An example of the power dissipation density calculated for the individual elements of one of the undeformed diabase textures (grid 1) is shown in Figure 5. This was calculated for an externally imposed strain rate of $10^{-4}s^{-1}$; it represents the scalar product of the stress and strain rate tensors corresponding to Figures 4*a* and 4*b*.

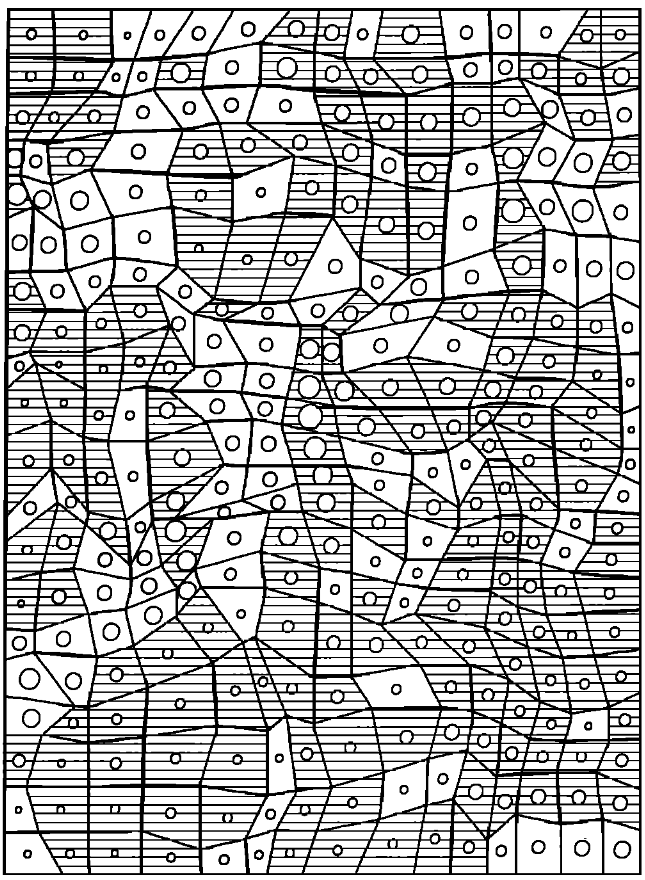

Fig. 5. Distribution of the power dissipation density (work per time per volume) for the same model diabase computation illustrated in Figures 4*a* and *b*. The pyroxene (shaded) is stronger than the plagioclase (unshaded) but dissipates less power per unit volume, as indicated by the larger circles in the unshaded areas.

Idealized Textures

We have done calculations for four idealized textures of the type shown in Figure 2, consisting of the four combinations of inclusions that are square or circular, strong or weak. For these four textures we have made model calculations (at all five strain rates) for volume fractions of pyroxene and plagioclase that are approximately equal to those of the diabase grids. We have also done calculations for a wide range of different volume proportions of the two-phases (10-90%, in 5% increments), using both pyroxene and plagioclase as the square inclusion. These calculations were done at strain rates of $10^{-12}s^{-1}$, the extreme values used in this study.

The aggregate strengths for the four idealized textures at five constant strain rates from 10^{-4} to $10^{-12}s^{-1}$, a temperature of 1000°C, and using the volume fractions of the diabase (64% pyroxene and 36% plagioclase) are shown in Figure 6*a*, on a log stress versus log strain rate plot. The results for one of the diabase textures (grid 2, Figure 1*a*) are included for comparison. Figure 6*b* shows an expanded representation of the results for the four idealized textures and those for all three diabase grids, so that the relative positions of the points can be more clearly seen. In this latter figure the model calculations for each strain rate are displayed in their correct relative position between the stress limits of the weaker and stronger end-member phases; the abscissa is the fraction of the stress value on a linear scale between the stress for the weaker phase (0.0) and that for the stronger phase (1.0). The actual stresses corresponding to the limits are given for each strain rate in the caption. Figure 6c shows an even more expanded representation of the results for the idealized textures and the diabase grids. In Figure 6*c* the stresses have been normalized between the uniform stress and uniform strain rate bounds.

The three diabase grids and the four idealized textures do not have precisely the same volume fractions of plagioclase and pyroxene. Their volume fractions differ from the nominal 0.36 pyroxene/0.64 plagioclase by as little as 0.0004 and as much as 0.0242. For representations of their steady state flow stress on a scale such as those used in Figures 3 or 6*a* the differences are almost too small to see. However, on expanded scales such as those of Figures 6*b* and 6*c*, the stress differences are evident. For this reason we have plotted a slightly adjusted stress in order to take these configurationally induced stress differences into account and to make the results more directly comparable from one geometry to another. Our procedure for computing the adjusted stresses is the following. We determine what fraction of the way the stress that we obtain from the finite element output is between the uniform stress and uniform strain rate

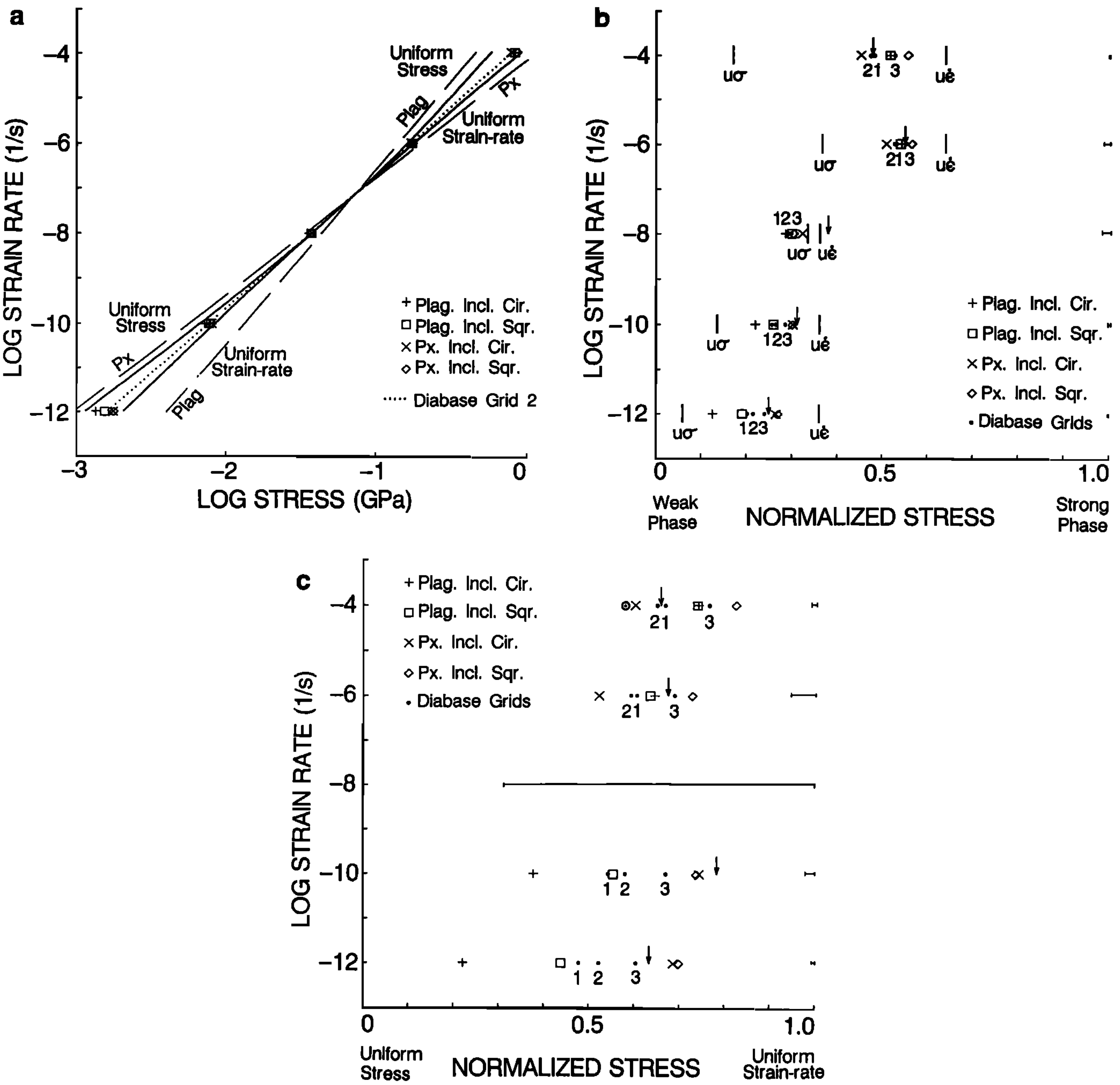

Fig. 6. (*a*) Plot showing the plane strain flow laws at 1000°C for the four idealized geometries. In each case the volume fractions are the same as for the three diabase grids of Figure 3. As in Figure 3, the experimental flow laws for pyroxene and plagioclase are shown as long-dashed lines; the uniform stress and uniform strain rate theoretical bounds are shown as the upper and lower solid lines, respectively; and the dotted line is the result for diabase grid 2. (*b*) An expanded plot showing the variation in stresses among the different models. The stress axis is normalized (by a zero shift and a scale change) between the limits of the flow strengths of the weak phase (left) and the strong phase (right) and is a linear, not a log, scale. See Figures 3 and 6*a* for comparison. The vertical bars show the uniform stress (uσ) and uniform strain rate (uε̇) bounds. The numbered points correspond to the three diabase grids. The results of a calculation using equations (10)-(12) are shown by the arrows. The small bars at the right of the figure show the stress magnitude associated with 0.5% accuracy for each strain rate. The bars are larger for cases of intermediate strain rates where the weak and strong phase strengths are similar, and the plot is therefore expanded more. The computational errors are evidently large enough that the computed stresses at $10^{-8}s^{-1}$ fall outside the theoretical bounds. The logs of the stresses (in gigapascals) corresponding to the weak-phase strength for volume fractions of 0.64 pyroxene and 0.36 plagioclase for log strain rates of -4, -6, -8, -10, and -12 are -0.340, -0.853, -1.470, -2.240, and -3.009. Those corresponding to the strong phase strengths (same strain rates) are 0.068, -0.701, -1.366, -1.878, and -2.391. The logs of the stresses corresponding to the uniform stress bound at the same strain rates are -0.238, -0.791, -1.433, -2.170, and -2.935. Those corresponding to the uniform strain rate bound are -0.039, -0.750, -1.430, -2.073, and -2.680. (*c*) A further expanded plot showing the variation in stress on a linear scale between the uniform stress and uniform strain rate bounds. The symbols have the same meaning as in Figure 6*b*. The open circle shows the behavior of the fine version of diabase grid 1 at $10^{-4}s^{-1}$. The error bars here also correspond to 0.5% accuracy on the stresses, and are longer due to the greatly expanded scale, especially at $10^{-8}s^{-1}$. The calculated stresses are not shown at that strain rate because they fall outside these bounds as shown in Figure 6*b*.

bounds for the actual volume fraction of that particular finite element example (these bounds are a function of volume fraction, as discussed below). This normalized stress is plotted in Figure 6*c* (and Figure 9*b*). The stresses plotted for Figures 6*b*, 6*a*, and 3 (and Figure 9*a*) are obtained by applying this normalization to the stresses for the uniform stress and uniform strain rate bounds that correspond to the nominal volume fractions of 0.36 (pyroxene) and 0.64 (plagioclase). This adjustment of the stresses is the best way to compare the results of the natural and idealized diabase textures with one another. The adjustments are insignificant if one only wants to know the stress for a given strain rate but become important for comparisons between different geometries. The adjustments involved at $\dot{\varepsilon} = 10^{-12}s^{-1}$ are the largest; for example, on Figure 6*a* the changes in the log stress range from 0.0002 to 0.0122, and on Figure 6*b* the changes in the normalized stress range from 0.0003 to 0.0273.

Examples of the output stresses, strain rates, and power dissipation densities for individual elements for the four idealized textures are shown in Figures 7 and 8, all for a strain rate of $10^{-12}s^{-1}$ for which plagioclase (unshaded) is the stronger phase. These figures allow one to compare the behaviors of weak and strong inclusions, as well as the behaviors of circular and square inclusions, at two different volume fractions.

The effect of the volume fraction of the inclusion on the aggregate strength is shown in Figure 9. Figure 9*a* shows a plot of weak phase volume fraction versus aggregate flow stress normalized between the plagioclase and pyroxene end-member flow stresses for the two strain rates. The variation of the uniform strain rate and uniform stress bounds with volume fraction are also shown. Figure 9*b* plots the same data normalized between the uniform stress and uniform strain rate bounds appropriate for each volume fraction. Figures 9*a* and 9*b* also provide a test of our finite element model calculations. They show that all of the volume fractions of square inclusions studied (10-90%) do fall between the uniform strain rate and uniform stress bounds.

DISCUSSION

In this section we discuss some of the important results from the finite element models of the diabase textures and of the idealized textures, including the effects that spatial configuration and volume fraction of the phases can have on the value of the aggregate strength within the uniform stress and uniform strain rate bounds. Then we discuss two useful methods that we have devised for simply calculating close approximations for the flow law of a polyphase aggregate from the flow laws and volume proportions of the phases. Finally, we discuss some applications to experimentally and naturally deformed polyphase aggregates.

Diabase Textures

An obvious test of the finite element model is to compare the calculated flow stresses with those measured experimentally by *Shelton* [1981] for Maryland diabase (the short-dashed line in Figure 3; dotted rectangle indicates experimental conditions). The general positions and slopes of the model and experimental flow laws are similar, although they do not agree in detail, and the experimental flow law does not pass through the equiviscous point associated with the input flow laws for the model. These disagreements may be due to differences between the pyroxene and plagioclase in the diabase and in the monomineralic aggregates.

Grids 1 and 2, fit to different regions of the undeformed diabase, give slightly different stresses at any given strain rate. As nearly as the eye can judge, the diabase is uniform in texture. Thus the slight differences in stress may be an indication that the area of our grid (roughly 50 grains) is not large enough to smooth out local heterogeneities. There are larger differences between the results for the deformed diabase grid and the two undeformed grids; these are discussed below.

One very important result which can be seen on Figure 3 is that although the log stress versus log strain rate curves for all three diabase grids are not exactly straight lines, their curvature is small. Thus even for this case where the stress exponents of the two constituent phases differ significantly (n = 3.9 and 2.6), the approximation of a simple power law for the polyphase aggregate is nearly correct.

The stress and strain rate distributions shown in Figure 4 show that neither homogeneous strain nor homogeneous stress is a good approximation in deformed polyphase aggregates. Our neglect of crystal anisotropy and our use of a relatively coarse grid compared to the sizes of the regions of the two different phases suggest that the inhomogeneities in real aggregates would be even greater. However, the possibilities for variations in strain rate perpendicular to the plane of the section in a real three-dimensional aggregate might tend to reduce these inhomogeneities compared to our two-dimensional model [e.g., *Karlsson and Linden*, 1975].

Comparisons of the power dissipation density between different "grains" of a given phase (Figure 5) show that details of the surroundings exert an important influence on the local behavior. The power dissipation density in the weaker phase is larger if it is part of a large weak area than if it occupies a small region surrounded by the stronger phase. Also, the power dissipation in the stronger phase is larger in small areas surrounded by the weak phase than in large strong areas. In general, the power dissipation density is similar for the two phases because it represents the scalar product of the stress and strain rates, and these vary antithetically for the two phases.

Effect of Inclusion Strength on Aggregate Strength

The artificial geometries that we studied (Figure 2) represent different cases of isolated inclusions in a continuous matrix. Whether the inclusion is the weaker phase or the stronger phase makes a difference in the behavior of the aggregate. It affects the proximity of the aggregate strength to the uniform strain rate or uniform stress bounds, as a result of differences in the distribution of stress and strain rate between the phases. The situations of weak and strong incusions will be discussed in turn.

For the case of weak inclusions the strain rate in the inclusion is similar to that for the aggregate as a whole (Figures 7*d* and 8*d*), and thus the approximation of uniform strain rate is reasonable. The aggregate strength varies between 0.7 and 0.9 of the way from the uniform stress to the uniform strain rate bound (Figure 9*b*). For weak inclusions the effect of volume fraction on aggregate strength, within the bounds, is minor. For square inclusions this is true over the entire range from 10 to 90% of the weak phase. Our limited results for circular inclusions as the weak phase (occupying 36 and 64% by volume) are consistent with those for square inclusions.

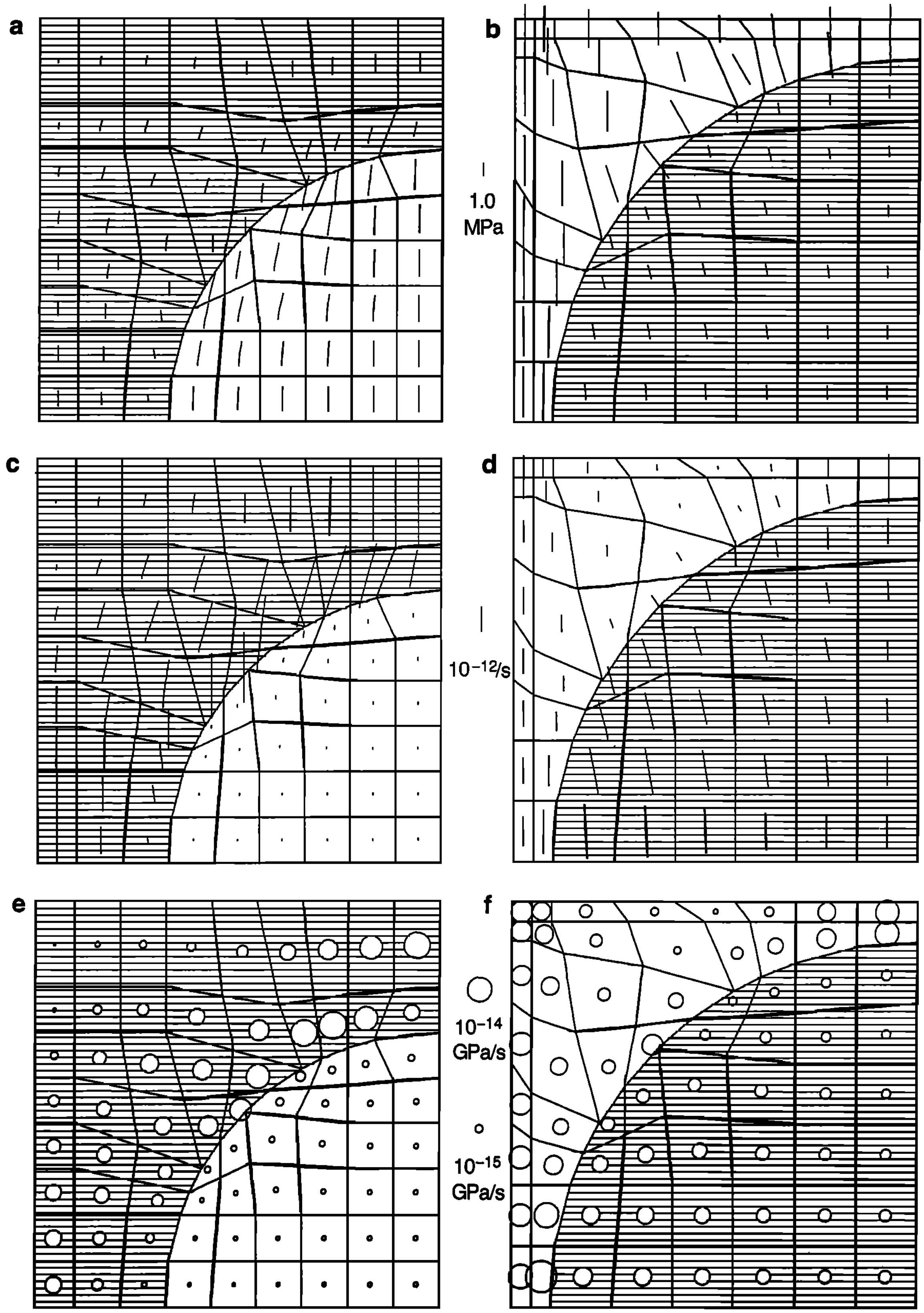

Fig. 7. Distribution of (*a* and *b*) stresses, (*c* and *d*) strain rates, and (*e* and *f*) power dissipation densities for the two idealized circular inclusions. The computations are for $10^{-12}s^{-1}$ and 1000°C, where plagioclase (unshaded) is the stronger phase. The size of the inclusions in these two geometries differs because in each case the plagioclase is 36% of the area. The significance of the symbols showing the stresses, strain rates, and power dissipation is the same as in Figures 4 and 5 for the diabase grid. The magnitudes of the stresses, strain rates, and power dissipation densities are much smaller than those in Figures 4 and 5 due to the much lower imposed average strain rate.

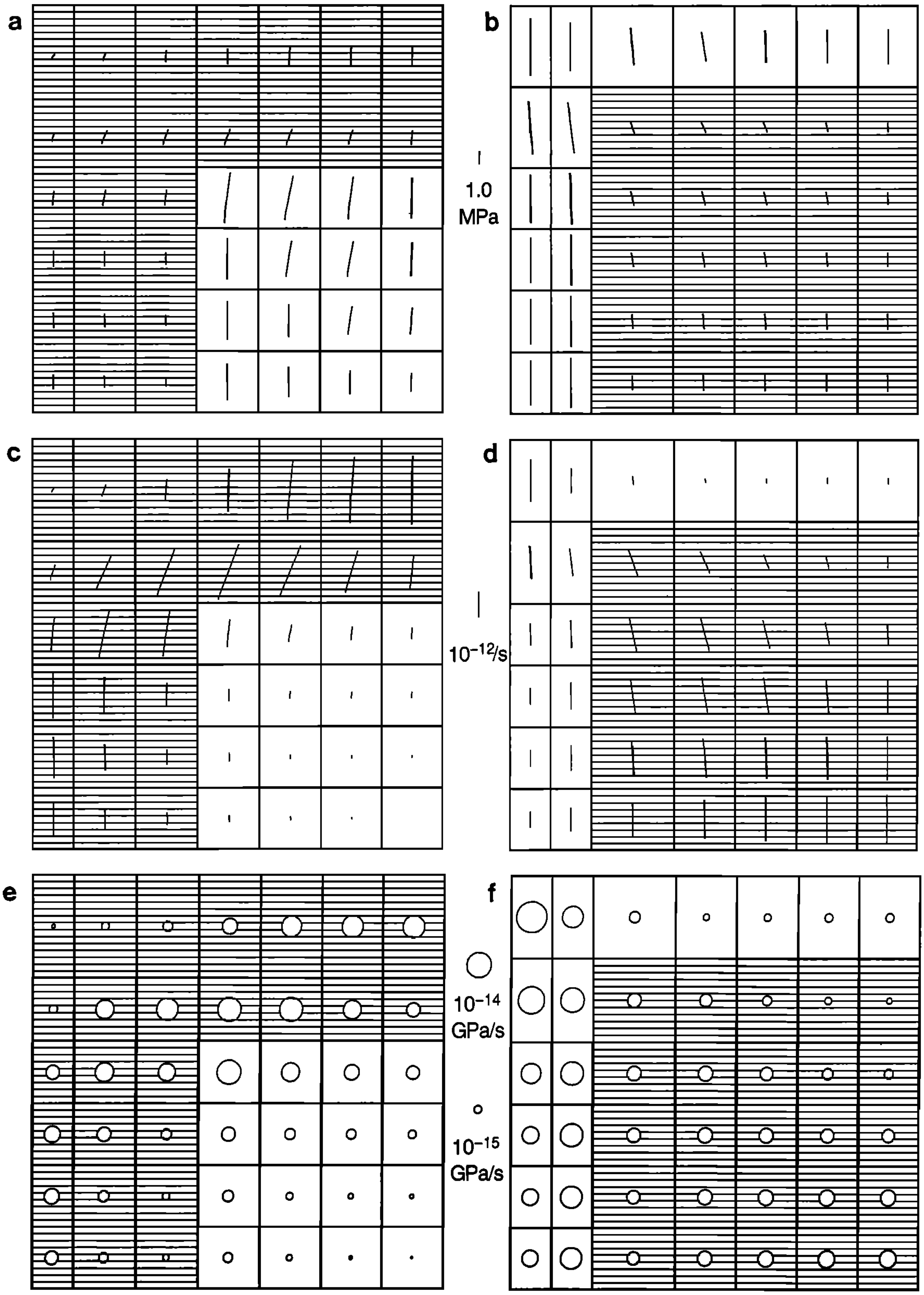

Fig. 8. Distribution of (*a* and *b*) stresses, (*c* and *d*) strain rates, and (*e* and *f*) power dissipation densities for the idealized square inclusions. The symbols and conditions are the same as those in Figure 7.

This is shown in Figure 9*b* and also can be seen in more detail in Figures 6*b* and 6*c* by comparing circular and square weak plagioclase inclusions at 10^{-4} and $10^{-6}s^{-1}$ or circular and square weak pyroxene inclusions at 10^{-10} and $10^{-12}s^{-1}$. The fact that weak inclusions result in aggregate behavior close to the uniform strain rate bound is an intuitively reasonable result that should have general applicability. If there is no connection between the weak inclusions, deformation of the aggregate must involve strain of the stronger matrix, and not much concentration of strain is possible in the inclusions.

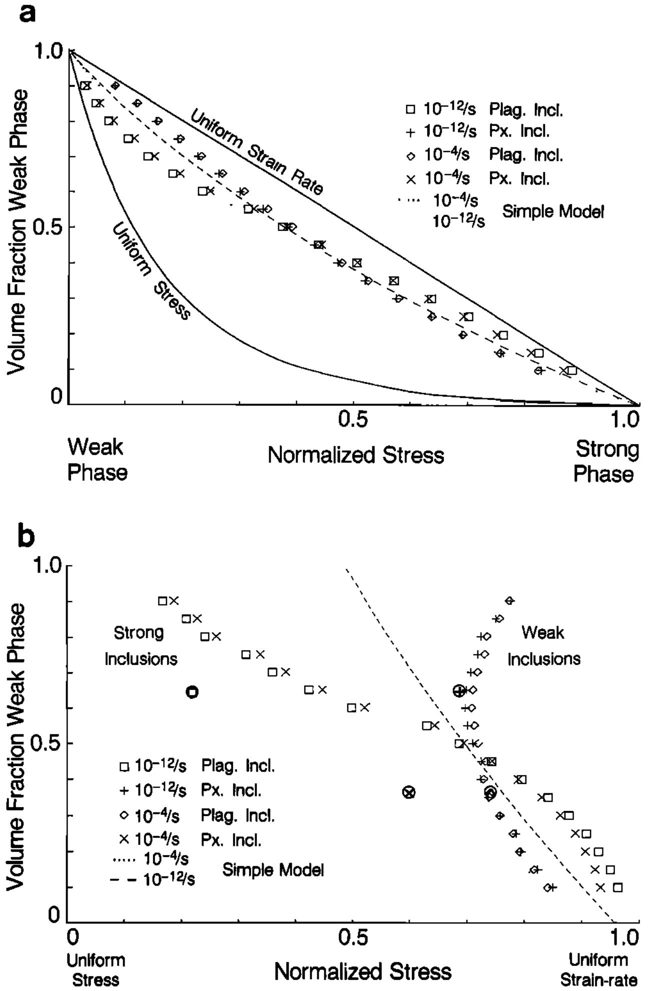

Fig. 9. Plots showing effect of volume fraction on the strength of a two-phase aggregate, for weak and for strong square-shaped inclusions at each of two different strain rates. The horizontal axes represent normalized stress on a linear scale either between the limits of the end-member or the theoretical bounding strengths. On both plots the points for the strong inclusions (plagioclase at $10^{-12}s^{-1}$ and pyroxene at $10^{-4}s^{-1}$) tend to plot together, as do those for the weak inclusions. (*a*) Plot of the aggregate strength normalized between those of the end-members, showing that volume fraction plays a major role in dictating the aggregate strength. (*b*) Plot showing the strengths normalized between the uniform stress and uniform strain rate bounds. The dotted and dashed lines show the predictions of the simple model discussed in the text. Points have been included showing the strengths of circular inclusions for volume fractions of 36 and 64%; these are denoted by standard symbols surrounded by circles.

Indeed, the strain rate is considerably more uniform in the aggregate when the inclusions are weak (Figures 7*d* and 8*d*) than when they are strong (Figures 7*c* and 8*c*).

The situation is more complicated for the case of strong inclusions, because the distributions of stress and strain rate depend on the volume fraction of the strong inclusion. For the square inclusion geometries the strength varies from about 0.2 to 0.98 of the way from the uniform stress to the uniform strain rate bound with increasing volume fraction of strong inclusion (Figure 9*b*). The two points for strong circular inclusions (constituting 64 and 36% by volume) plot about 0.2 closer to the uniform stress bounds than do the points for the same volume fraction with strong square inclusions (Figure 9*b*; also see Figure 6*c*, plagioclase at 10^{-10} and $10^{-12}s^{-1}$ and pyroxene at 10^{-4} and $10^{-6}s^{-1}$). This means that the difference between the behavior of weak and strong low-volume fraction inclusions is even greater for circular inclusions than for the square ones illustrated in Figure 9. This effect of shape is discussed below.

Why does the volume fraction of strong inclusions affect the aggregate behavior beyond influencing the values of the uniform stress and strain rate bounds? The case of low-volume fractions of a strong inclusion is similar to a stiff inclusion in an infinite medium, a situation in linear elasticity where the stresses are known to be similar in both inclusion and matrix [*Jaeger and Cook*, 1969, pp. 248-251]. The fact that this result seems to be true for both linear elastic strain and nonlinear plastic flow suggests that it is of general validity. It is an intuitively reasonable result; an isolated strong inclusion cannot act as a significant stress riser because the stress must be transmitted to it through the weaker matrix, and the distance between strong inclusions is large enough that stress concentrations should not be produced in the matrix. However, as the volume fraction of the strong inclusions is increased, the smaller volume of weak matrix is forced to flow faster in the decreased space between the inclusions. Thus the stress in the matrix will become less uniform, and so it will load the inclusions less uniformly. For high-volume fractions of strong inclusions, load will be shifted to the strong inclusions, and the strain will be nearly uniform between the phases. Although it might seem that the small amount of weak matrix between the strong inclusions could strain more than the strong inclusion in the y direction (see Figure 2), the compensating strain of opposite sign in the x direction must be similar in both phases, and this restricts them to relatively similar strains.

Effect of Inclusion Shape on Aggregate Strength

If the inclusions are strong, their shape has a definite effect on the strength of the aggregate: square inclusions provide a greater contribution to the aggregate strength than circular inclusions. This can be seen in Figure 9 and in Figures 6*b* and 6*c* by comparing circular and square plagioclase inclusions at 10^{-10} and $10^{-12}s^{-1}$, and circular and square pyroxene inclusions at 10^{-4} and $10^{-6}s^{-1}$. The explanation for this ultimately lies in the greater stress concentrations in the square inclusion where its corners interfere with the flow in the surrounding matrix (compare Figures 8*a* and 8*c* with 7*a* and 7*c*). This results in a higher average power dissipation density in square strong inclusions than in circular ones (compare Figures 8*e* and 7*e*). The shape which results in a greater power dissipation will have a greater influence on the overall aggregate flow behavior. Consequently, the strong inclusions contribute more to the strength when they are square, and they cause the aggregate strength to be close to the high stress (uniform strain rate) bound. When the strong inclusions are circular, the aggregate strenth is closer to the low stress (uniform stress) bound.

In contrast to the situation for strong inclusions, the average power dissipation density for weak inclusions is not nearly so dependent on the shape of the inclusions (see Figures 7*f* and 8*f*). This is because a weak inclusion tends to deform about the same amount as the surrounding stronger matrix, so stress and strain rate concentrations do not develop for square or circular inclusions.

Comparison of Results for Diabase and Idealized Textures

Some of the details of the finite element results for diabase can be better understood in terms of the results for the idealized geometries. The discussion of the idealized geometries shows that the details of the phase configuration are more important for situations where the stronger phase occurs as isolated inclusions. The results for all three diabase grids and for the four idealized textures with the same volume proportions (36% plagioclase, 64% pyroxene) show similar behavior (Figure 6*c*): on the high strain rate side of the equiviscous point the results group together about 70% of the way toward the uniform strain rate bound, whereas below the equiviscous point they tend to be more centered between the uniform stress and uniform strain rate bounds. We believe that this result is due to the unequal volume fractions of the two phases in the diabase.

For diabase on the high strain rate side of the equiviscous point the smaller volume phase (plagioclase) is weaker. Our discussion of the results for the various idealized phase configurations (Figure 9*b*) suggests that the strength of such an aggregate should be about 75% of the way toward the low stress bound, and indeed it is.

For diabase on the low strain rate side of the equiviscous point the smaller volume phase is stronger. Two of the idealized texture cases have perfectly isolated strong inclusions, and two have a continuous matrix of the strong phase. The relative positions of the aggregate strengths below the equiviscous point (Figure 6*c*) are as follows: the cases of perfectly isolated inclusions of strong phase are closer to the uniform stress bound, the cases of perfectly continuous strong matrix are closer to the uniform strain rate bound, and the three diabase grids (partly interconnected strong phase) fall in between. This explains why there is more variability in the diabase strengths below the equiviscous point than above it: the geometric arrangements are more important if the smaller volume fraction is the stronger phase.

It is noteworthy that diabase grids 1 and 2, taken from the undeformed samples, group together on Figures 6*c* and 6*b*, and that diabase grid 3, taken from a deformed sample, is always somewhat stronger. We believe this results from the fact the deformatiuon has produced layering of the phases, with greater connectivity in the elongation direction than existed in the undeformed state (see Figure 1). This means that it is harder for the aggregate to deform by shortening perpendicular to the foliation and elongation parallel to it, regardless of which phase is the stronger, because deformation of the aggregate requires greater deformation of the stronger phase than would be true for the textures of the undeformed samples. In aggregates where the initial phase configuration involved an interconnected strong phase and a dispersed weak phase, deformation may result in a reversal of the phase connectivity and thus a weakening of the aggregate. This has been observed experimentally in aplite [*Dell'Angelo and Tullis*, 1982] and inferred in many naturally deformed polyphase aggregates [*Handy*, 1990], but the diabase does not show this textural evolution.

Estimation of Polyphase Aggregate Flow Law From End-Member Flow Laws

The finite element modeling is useful for accurate evaluations of the contributions of different factors to the flow law of a polyphase aggregate. However, it is too tedious to employ for each new aggregate. We have developed two relatively simple and surprisingly accurate procedures for approximating the flow law of polyphase aggregates, based on the flow laws of the constituent phases, which we find to agree well with the finite element models for the diabase grids and idealized geometries that we have considered. One procedure can be used when the volume fractions and phase configurations are well known and involves use of the uniform stress and uniform strain rate bounds. The other can be used when only the volume fractions are known; this procedure gives the aggregate flow law in the form of equation (1).

For all phase configurations our finite element calculations show that the flow law for a polyphase aggregate can be closely approximated as a power law, in spite of the fact that this is not rigorously true if the constituent phases have different stress exponents. Thus the aggregate flow law can be approximated as a straight line on a log $\dot{\varepsilon}$ versus log σ plot, and this straight line must pass through the equiviscous point where the power laws for the two end-members intersect. The slope n (equal to the effective stress exponent) of this aggregate power law is therefore the only additional parameter needed to characterize it fully. The slope must lie between the limits of the end-member slopes, and it must also lie between the uniform stress and uniform strain rate bounds for the volume fractions of interest. These bounds shift with volume fractions so that as the volume fraction of a given end-member increases, both bounds become closer to the straight line representing the power law for that end-member.

If the volume fractions and phase configurations for the aggregate of interest are well known, then a reasonably accurate value for the aggregate strength can be obtained using the results presented in the previous sections, which we summarize here in the form of the following four rules. (1) For inclusions that are weaker than the matrix the position of the aggregate strength between the uniform stress and uniform strain rate bounds does not depend very much on the volume proportion of the phases and averages about 0.78 of the way toward the uniform strain rate (high stress) bound. (Note that this is 0.78 of the stresses, not of the log stresses.) (2) For inclusions that are stronger than the matrix the position between the bounds varies approximately linearly with the volume proportion of strong phase from the uniform stress (low stress) bound to the uniform strain rate bound. (3) If the configuration of the phases cannot be described as inclusions of one phase in the other but instead the phases are both interconnected, then the behavior tends to fall somewhere between the cases for weak and strong inclusions. (4) Phase distributions that allow nearly all strain to be taken up by shearing in one phase will have strengths that fall near the uniform stress bound.

These rules can be used to estimate the strength of the aggregate as a function of strain rate, using the following procedure. At several strain rates, calculate the uniform $\dot{\varepsilon}$ and the uniform σ bounds. Then estimate the strength using the above four rules. A plot of these results forms a curve which represents the estimated flow law. When we have tried this approach for various imagined volume fractions and configurations, we have found that the resulting flow law is close to following a straight line on a log strain rate versus log stress plot.

We have found an even simpler procedure for approximating the aggregate flow law parameters n_a, Q_a, and A_a (equation (1)) which seems to work remarkably well. It can be used even if the phase configurations are not known, although it is not appropriate for situations when a weak phase is lined up along shear planes. The log of the aggregate stress exponent is found to fall between the logs of the end-member stress exponents in proportion to the volume fraction of the end-members in the aggregate. Thus, if the aggregate consists of 36% plagioclase by volume, the log of the stress exponent for the aggregate is 0.36 of the way from the log of the stress exponent of pyroxene to that of plagioclase. We have no rigorous justification for this procedure, although it makes qualitative sense due to the way in which the uniform stress and uniform strain rate bounds shift with the volume fractions of the phases. We have found the procedure to work well for several idealized geometries and volume fractions that we tested, when compared with the procedure outlined above. The fact that the linear proportion between the logs of the stress exponents gives a better match to the finite element predictions than a linear proportion between the stress exponents themselves amounts to using a geometric proportion rather than an arithmetic one. The geometric mean rather than the arithmetic mean of the slopes of two lines will yield a line that is symmetrically disposed between them on a plot whose axes are scaled so that the angle between one line and either coordinate axis equals the angle between the other line and the other coordinate axis.

In sum, we propose that the flow law for a two-phase aggregate can be approximated by a power law that passes through the intersection point of the end-member flow laws at a given temperature and for which the power law exponent is the volume-weighted geometric mean of the end-member exponents. This yields the following three equations for the aggregate power law parameters:

$$n_a = 10^{(f_1 \log n_1 + f_2 \log n_2)} \tag{10}$$

$$Q_a = \frac{Q_2\,(n_a - n_1) - Q_1\,(n_a - n_2)}{(n_2 - n_1)} \tag{11}$$

$$A_a = 10^{[\log A_2 (n_a - n_1) - \log A_1 (n_a - n_2)] / [n_2 - n_1]} \tag{12}$$

where n, Q, and A represent the power law exponents, activation enthalpies, and pre-exponential factors in flow laws of the form of equation (1); the subscripts a, 1, and 2 stand for the aggregate, phase 1, and phase 2, respectively; and f represents volume fraction.

An illustration of polyphase flow laws calculated using the procedure just described is shown in Figures 6*b* and 6*c*, where an arrow indicates the stress calculated using this procedure for each of the strain rates. The arrow typically approximates an average of the various diabase grids and idealized geometries studied. It represents particularly well the average of the diabase grids, but it is not as satisfactory in fitting the extreme cases represented by the strong inclusions. The prediction of the simple model is also shown on Figure 9, where it is seen to approximate the results found for different volume fractions using the finite element method.

Equations corresponding to (10), (11), and (12) could be obtained for a polyphase aggregate consisting of any number of phases. Generalizing equation (10) to obtain the slope n_a for an aggregate containing m phases is straightforward:

$$n_a = 10^{(\sum_{i=1}^{m} f_i \log n_i)} \tag{13}$$

It is less obvious how to generalize equations (11) and (12), and we have not obtained expressions for these. However, if one can determine the appropriate point on a log $\dot{\varepsilon}$ versus log σ plot through which the line of slope n should pass, then a line representing the strength of the polyphase aggregate can be drawn for any temperature of interest. We suggest the point be found as follows. First, draw the m straight lines corresponding to the m phases. These will have $m!/(m-2)!2!$ intersection points. The point of interest is the weighted center of gravity of these points, the weighting factor being the average of the volume fractions of the two phases corresponding to each intersection point.

Comparison With Other Methods

An alternative model for deformation of two-phase aggregates was developed by *Tharp* [1984] based on analogy with porous metal composites, and it is of interest to compare his results with ours. He considered that porous powder metals might be a good approximation for two-phase aggregates deforming by power law creep, if one phase is 1-2 orders of magnitude weaker than the other. Using such a model, he found that when the weaker phase constitutes 15%, the aggregate flow stress is reduced by 50%, and when it constitutes 40%, the flow stress is reduced by an order of magnitude. Our equations (10)-(12) agree with his results only if the weaker phase is 2 orders of magnitude weaker. If it is only 1 order of magnitude weaker, our equations give an aggregate strength that is reduced by only 30% for 15% weak phase, and by 50% for 40% weak phase. Our predictions match the experimental results of *Jordan* [1987] on halite-calcite aggregates, for which the contrast in strength is about 1 order of magnitude.

A useful comparison can be made with the modeling of *Chen and Argon* [1979]. They assume that the strain in their spherical or circular inclusions is homogeneous, even though this is not necessarily known to be the case for two power law materials. Our Figures 8*c* and 8*d* show that the strain in our circular inclusions is relatively homogeneous for the grid size we used, and consequently their assumption seems reasonable.

CONCLUSIONS

In this study of the deformation of an aggregate comprising two power law materials, we have used a finite element method to investigate how the aggregate flow law depends on the flow laws, volume proportions, and configurations of the constituent phases and to investigate the spatial variation of the stresses and strain rates in the constituent phases. These results have allowed us to determine two methods for estimating the aggregate flow law even when the phase configuration is not known; the results of these methods match the finite element calculations well.

Our finite element model calculations show that the diabase flow strengths deviate only slightly from a straight-line extrapolation of the experimentally determined power law flow law. The quality of the approximation decreases with increasing difference between the stress exponents of the end-member phases. However, the stress exponents for the two materials used in this study are relatively far apart (3.9 and 2.6), which indicates that most polyphase aggregates in which both phases deform by dislocation creep can be quite accurately approximated as simple power law materials.

The most important factor controlling the strength of an aggregate is the volume proportion of the constituent phases. The aggregate strength must lie between the theoretical bounds of uniform stress and uniform strain rate, and the position of these bounds is a function of the volume fraction, shifting toward the volumetrically more abundant end-member. Calculation of these bounds is relatively simple and is one useful approach to delimiting the aggregate strength. For phase geometries approximated by simple shear parallel to compositional layering or by a small proportion of strong inclusions in a weak matrix, the aggregate strength will be close to the uniform stress bound.

For most other phase geometries we have found that the flow law of a polyphase aggregate can be closely approximated by a simple method. This method approximates the aggregate flow law as a power law that passes through the intersection point of the end-member flow laws; the power law exponent is the volume-weighted geometric mean of the end-member exponents. The stress exponent, activation enthalpy, and pre-exponential factor of the aggregate flow law are given by equations (10)-(12). Thus if one knows the flow laws for a relatively small number of monomineralic aggregates (quartz, feldspar, pyroxene, olivine), one can calculate flow laws for large portions of the lower crust and upper mantle.

Acknowledgments. This work was supported by USGS contract 14-08-0001-15247 (J.T. and T.T.) and NSF grants EAR-8708356 (J.T.) and EAR-8816791 (T.T.). We are grateful to A. Needleman for his generosity in supplying the initial program which we modified for this project and for helpful discussions, especially on convergence criteria. We are also grateful to A. Kronenberg for helpful discussions, and to T. Tharp, M. Handy, A. Kronenberg, and an anonymous reviewer for useful and constructive reviews.

REFERENCES

Ave'Lallemant, H. G., and N. L. Carter, Syntectonic recrystallization of olivine and modes of flow in the upper mantle, *Geol. Soc. Am. Bull.*, *81*, 2203-2220, 1970.

Chen, I. W., and A. S. Argon, Steady state power law creep in heterogeneous alloys with coarse microstructures, *Acta Metall.*, *27*, 785-791, 1979.

Chen, I. W., and W. P. Molnar, Focal depths of intracontinental and intraplate earthquakes and its implications for the thermal and mechanical properties of the lithosphere, *J. Geophys. Res.*, *88*, 4183-4214, 1983.

Christensen, U. R., Convection with temperature and pressure dependent non Newtonian rheology, *Geophys. J. R. Astron. Soc.*, *77*, 343-384, 1984.

Dell'Angelo, L., and J. Tullis, Textural strain softening in experimentally deformed aplite, *Eos Trans. AGU*, *63*, 438, 1982.

Gilotti, J. A., The rheologically critical matrix percentage in quartzofeldspathic mylonites, in *Structural Geology of Fold and Thrust Belts*, Johns Hopkins University Press, Baltimore, Md. in press, 1991.

Green, H. G., and R. S. Borch, A new molten salt cell for precision stress measurement at high pressure, *Eur. J. Mineral.*, *1*, 213-219, 1989.

Handy, M. R., The solid-state flow of polymineralic rocks, *J. Geophys. Res.*, *95*, 8647-8661, 1990.

Hashin, Z., Theory of mechanical behavior of heterogeneous media, *Appl. Mech. Rev.*, *17*, 1-9, 1964.

Hashin, Z., and S. Shtrikman, A variational approach to the theory of the elastic behavior of multiphase materials, *J. Mech. Phys. Solids*, *11*, 127-140, 1963.

Heard, H. C. and C. B. Raleigh, Steady state flow in marble at 500° to 800°C, *Geol. Soc. Am. Bull.*, *83*, 935-956, 1972.

Hill, R., A self-consistent mechanics of composite materials, *J. Mech. Phys. Solids*, *13*, 213-222, 1965.

Horowitz, F. G., T. E. Tullis, A. K. Kronenberg, J. Tullis, and A. Needleman, Finite element model of polyphase flow, *Eos Trans. AGU*, *62*, 396-397, 1981.

Jaeger, J. C., and N. G. W. Cook, *Fundamentals of Rock Mechanics*, 513 pp., Methuen, New York, 1969.

Jordan, P. G., The deformational behavior of bimineralic limestone-halite aggregates, *Tectonophysics*, *135*, 185-197, 1987.

Jordan, P. G., The rheology of polymineralic rocks – An approach, *Geol. Rundsch.*, *77*, 285-294, 1988.

Karlsson, B., and G. Linden, Plastic deformation of ferrite-pearlite structures in steel, *Mater. Sci. Eng.*, *17*, 209-219, 1975.

Kronenberg, A. K., and G. L. Shelton, Deformation microstructures in experimentally deformed Maryland diabase, *J. Struct. Geol.*, *2*, 341-353, 1980.

LeHazif, R., Deformation plastique du systeme biphase fer-argent de composition equivolume, *Acta Metall.*, *26*, 247-257, 1978.

Needleman, A., and C. F. Shih, A finite element method for plane strain deformations of incompressible solids, *Computer Methods in Appl. Mech. and Eng.*, *15*, 223-240, 1978.

Nicolas, A. and J. P. Poirier, *Crystalline Plasticity and Solid State Flow in Metamorphic Rocks*, 444 pp., John Wiley, New York, 1976.

Nye, J. F., The flow law of ice from measurements in glacier tunnels, laboratory experiments and the Jungfraufirn borehole experiment, *Proc. Roy. Soc. London, Ser. A*, *219*, 477-489, 1953.

Paterson, M. S., Problems in the extrapolation of laboratory rheological data, *Tectonophysics*, *133*, 33-43, 1987.

Price, R. H., Effects of anhydrite and pressure on the mechanical behavior of synthetic rocksalt, *Geophys. Res. Lett.*, *9*, 1029-1032, 1982.

Ross J. V., S. J. Bauer, and F. D. Hansen, Textural evolution of synthetic anhydrite-halite mylonites, *Tectonophysics.*, *140*, 307-326, 1987.

Shelton, G. L., Experimental deformation of single and polyphase crustal rocks at high temperatures and pressures, Ph.D. thesis, 146 pp., Brown Univ., Providence, R. I., 1981.

Shelton, G. L., and J. Tullis, Experimental flow laws for crustal rocks, *Eos Trans. AGU*, *62*, 396, 1981.

Sibson, R.H., Fault zone models, heat flow, and the depth distribution of earthquakes in the continental crust of the United States, *Bull. Seismol. Soc. Am.*, *72*, 151-163, 1982.

Stocker, R. L. and M. F. Ashby, On the rheology of the upper mantle, *Rev. Geophys.*, *11*, 391-426, 1973.

Tharp, T., Analogies between the high-temperature deformation of polyphase rocks and the mechanical behavior of porous powder metal, *Tectonophysics*, *96*, T1-T11, 1984.

Tullis, J., and D. Mardon, Effect of muscovite content on deformation of quartz aggregates, *Geol. Soc. Am. Abstr. Programs.*, *16*, 678, 1984.

Tullis, J. and R. A. Yund, Hydrolytic weakening of quartz aggregates: The role of 'water' and pressure on recovery, *Geophys. Res. Lett.*, *16*, 1343-1346, 1989.

Zuber, M. T., E. M. Parmentier, and R. C. Fletcher, Extension of continental lithosphere: A model for two scales of Basin and Range deformation, *J. Geophys. Res.*, *91*, 4826-4838, 1986.

T. E. Tullis and J. Tullis, Department of Geological Sciences, Brown University, Providence, RI 02912.

F. G. Horowitz, Department of Civil Engineering, Northwestern University, Evanston, IL 60201.

(Received June 19, 1990;
revised October 29, 1990;
accepted November 14, 1990.)